

An Itô process that is not a martingale

[Thoughts dump]

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Fix a process f that is almost surely square-integrable. We know that the Itô integral $M_t := \int_0^t f_s dW_s$ is a local martingale. The question is, how can it be that M_t is not a true martingale? It is a martingale if $\int f^2 ds$ has a finite expectation.

Let $dR = \frac{1}{R} dt + dW$ with $R(0) = 1$ and consider the process $1/R$. By Itô formula,

$$1/R_t = 1 + \int_0^t -\frac{1}{R_s^2} dW_s$$

When R gets close to zero, $1/R$ explodes. $1/R$ is a local martingale as an Itô integral, but is it a martingale? Stop $1/R$ when it hits the level n :

$$\tau_n := \inf_{R_t \leq 1/n} t.$$

The stopped process $(1/R)_{t \wedge \tau_n} = 1 - \int_0^{t \wedge \tau_n} \frac{1}{R_s^2} dW_s$ is then a martingale because the integrand $\frac{1}{R_s^2}$ is bounded and so $\int_0^t (R_s^{-2})^2 ds$ has a finite expectation.

What breaks with the non-stopped process $1/R_t$?

It has always seemed to me that if a process is a martingale “up to a point”, and that “point” can be arbitrary far into the future, then surely the process is a martingale as a whole towards infinity?

Moreover, the stopped processes “obviously” converge to the unstopped one almost surely for each $t \geq 0$:

$$(1/R)_{t \wedge \tau_n} \xrightarrow{n \rightarrow \infty} (1/R)_t,$$

as a consequence of the stopping times diverging to infinity almost surely $\tau_n \rightarrow \infty$.

Now we would like to take $\mathbb{E}$ on both sides. Because each stopped process is a martingale, its mean is

$$\mathbb{E}(1/R_{t \wedge \tau_n}) = 1, \quad \text{for all } n.$$

But if we let n tend to infinity and so $1/R_{t \wedge \tau_n}$ reaches $1/R_t$ almost surely, we have no guarantee that the expectation preserves that limit.

A great counterexample is the function sequence $X_n(\omega) = n\mathbf{1}_{(0,1/n)}$ on $\Omega = [0, 1]$. On each ω , $X_n(\omega)$ tends to zero, so $X_n \rightarrow 0$ almost surely. But while all the functions have a unit mean $\mathbb{E}X_n = 1$, the limit is the absolute zero. Almost sure convergence is not strong enough to carry the $\mathbb{E}$ in the limit, as the functions X_n are explodingly tall on small but non-neglectible sets $(0, 1/n)$.

A similar thing is happening here. Let us decompose the stopped processes into their explosive and nice parts:

$$\begin{aligned} 1/R_{t \wedge \tau_n} &= 1/R_t \cdot \mathbf{1}_{t \leq \tau_n} + 1/R_{\tau_n} \cdot \mathbf{1}_{t > \tau_n} \\ &= 1/R_t \cdot \mathbf{1}_{t \leq \tau_n} + n\mathbf{1}_{t > \tau_n} \end{aligned}$$

and take expectations

$$1 = \mathbb{E}(1/R_t \cdot \mathbf{1}_{t \leq \tau_n}) + n\mathbb{P}(t > \tau_n).$$

If we let n to infinity in that decomposition, the truncated $1/R_t \cdot \mathbf{1}_{t \leq \tau_n}$ converges almost surely to $1/R_t$ *monotonically*, and so we can take the limit inside the expectation: $\mathbb{E}(1/R_t \cdot \mathbf{1}_{t \leq \tau_n}) \rightarrow \mathbb{E}(1/R_t)$. So we get a tangible expression for the limit expectation

$$\mathbb{E}(1/R_t) = 1 - \lim_n n \cdot \mathbb{P}(t > \tau_n).$$

Here Chatgpt says that $\mathbb{P}(t > \tau_n)$ does not die out faster than n so the limit is nonzero.

So our stopped processes are like the $n\mathbf{1}_{(0,1/n)}$ functions – they converge to their unstopped father, but only ω -pointwise, with small parts of Ω on which the stopped $1/R_t$ explode exactly so that their average over the whole of Ω is 1.